

```
import cv2
from google.colab import files
from google.colab.patches import cv2_imshow

# Upload image
uploaded = files.upload()

# Get image name
filename = list(uploaded.keys())[0]

# Read image in grayscale
img = cv2.imread(filename, 0)

# Apply Sobel operator along Y-axis
sobel_y = cv2.Sobel(img, cv2.CV_64F, 0, 1, ksize=5)

# Convert to 8-bit image
sobel_y = cv2.convertScaleAbs(sobel_y)

# Save output
cv2.imwrite("sobel_y.jpg", sobel_y)

# Display result
print("Original Image:")
cv2_imshow(img)

print("Sobel Y Edge Detection:")
cv2_imshow(sobel_y)

print("Output saved as sobel_y.jpg")
```

[Choose files](#) img-2.jpeg

img-2.jpeg(image/jpeg) - 8224 bytes, last modified: 20/07/2026 - 100% done

Saving img-2.jpeg to img-2.jpeg

Original Image:



Sobel Y Edge Detection:



Output saved as sobel_y.jpg